

SOUTHWEST MINNESOTA, MN, USA

WMO: 726559

Lat: 44.450N

Lon: 95.817W

Elev: 359

StdP: 97.08

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94976

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.4	(c) -22.4	(d) -29.9	(e) 0.2	(f) -23.5	(g) -27.3	(h) 0.3	(i) -21.8	(j) 16.1	(k) -11.5	(l) 14.7	(m) -9.3	(n) 4.9	(o) 320	(p) 0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.4	22.9	31.0	22.6	29.0	21.3	25.0	30.3	23.8	28.8	22.7	27.5	6.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.5	28.4	22.2	17.7	27.6	21.1	16.4	26.5	78.0	30.3	73.1	28.5	68.6	27.4	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.9	(b) 12.1	(c) 11.0	(e) -27.6	(f) 35.4	(g) 3.0	(h) 1.6	(i) -29.8	(j) 36.6	(k) -31.6	(l) 37.5	(m) -33.2	(n) 38.4	(o) -35.4	(p) 39.5		
(4) 13.9	(5) 12.1	(6) 11.0	(7) -27.6	(8) 35.4	(9) 3.0	(10) 1.6	(11) -29.8	(12) 36.6	(13) -31.6	(14) 37.5	(15) -33.2	(16) 38.4	(17) -35.4	(18) 39.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-9.2	-7.0	-0.3	7.2	14.3	20.1	22.4	20.8	16.8	9.2	1.0	-6.8
	DBStd	12.55	7.52	7.42	7.32	5.80	5.14	3.92	3.40	3.18	4.90	5.45	6.48	6.94
	HDD10.0	2416	595	477	325	118	18	0	0	0	5	81	277	520
	HDD18.3	4336	853	711	577	335	144	26	6	13	86	286	520	778
	CDD10.0	1486	0	0	7	35	153	302	383	333	209	57	7	0
	CDD18.3	363	0	0	1	2	21	78	131	88	41	4	0	0
	CDH23.3	3416	0	0	2	31	247	753	1251	711	374	46	1	0
	CDH26.7	1083	0	0	0	7	82	248	436	193	107	10	0	0
Wind	WSAvg	5.2	5.6	5.8	5.7	5.9	5.7	4.9	4.1	3.9	4.7	5.4	5.6	5.5
Precipitation	PrecAvg	714	20	17	38	72	86	111	84	91	82	59	34	21
	PrecMax	1006	54	38	89	191	169	279	158	171	307	164	107	59
	PrecMin	556	1	1	5	8	26	44	11	22	14	6	0	1
	PrecStd	125	15	10	24	39	44	50	38	40	66	44	32	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	13.2	22.1	27.3	32.7	33.8	34.9	32.5	32.3	27.9	21.0	12.0
		MCWB	3.9	9.0	13.1	15.9	19.3	21.8	25.5	24.6	21.8	16.7	12.2	7.6
	2%	DB	4.5	8.0	17.2	22.8	28.8	31.4	32.5	31.0	29.4	23.8	16.5	7.2
		MCWB	1.7	4.8	11.6	13.3	18.0	21.2	24.2	24.0	21.1	15.1	10.0	3.3
	5%	DB	2.7	5.2	12.8	19.7	26.2	29.1	31.1	28.7	27.5	21.0	12.9	3.9
		MCWB	0.6	2.3	8.6	11.7	16.3	20.0	23.7	22.6	19.3	13.7	7.9	1.3
	10%	DB	1.2	2.7	9.7	17.2	22.9	27.6	28.9	27.3	25.3	17.8	10.1	2.3
		MCWB	-0.3	0.8	6.6	10.2	14.8	18.9	22.4	21.0	18.1	11.8	6.0	0.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	8.8	15.1	16.8	21.8	24.7	27.1	26.1	23.6	19.1	13.9	7.7
		MCDB	8.0	13.3	18.8	25.1	29.7	30.2	31.9	30.9	29.4	25.2	18.5	10.8
	2%	WB	2.0	4.9	11.9	14.5	19.5	23.0	25.5	24.5	22.1	16.4	10.9	3.6
		MCDB	4.3	8.1	15.5	21.1	26.4	29.0	31.2	29.3	27.6	21.8	15.4	6.4
	5%	WB	0.9	2.7	9.4	12.6	17.8	21.5	24.2	23.3	20.7	14.4	8.3	1.6
		MCDB	2.7	5.1	13.3	18.3	24.2	27.8	29.4	27.7	25.7	19.7	13.1	4.0
	10%	WB	-0.4	0.9	6.4	10.8	16.1	20.1	23.1	21.9	19.3	12.4	6.0	0.2
		MCDB	1.1	2.9	9.6	16.1	21.4	26.2	27.8	26.2	24.1	17.3	9.8	2.1
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.2	9.2	11.0	11.3	10.7	10.8	10.7	12.0	10.8	9.2	8.3
		MCDBR	10.3	11.2	13.5	16.6	15.4	13.1	11.9	11.9	13.8	14.9	13.6	10.5
		MCWBR	8.4	8.4	9.2	9.6	7.9	6.5	6.3	6.7	7.1	8.4	9.0	8.1
	5% WB	MCDBR	10.0	10.8	13.2	14.2	13.0	11.8	11.1	10.9	11.8	13.0	12.8	10.0
		MCWBR	8.2	8.3	9.2	8.9	7.5	6.7	6.4	6.8	6.9	8.1	8.8	7.9
Clear-Sky Solar Irradiance	taub	0.263	0.276	0.309	0.362	0.381	0.395	0.404	0.408	0.365	0.322	0.286	0.267	
	taud	2.455	2.430	2.438	2.328	2.344	2.364	2.324	2.290	2.388	2.493	2.516	2.456	
	Ebn at Noon	877	925	931	900	889	875	861	844	857	856	841	832	
	Edh at Noon	70	86	98	120	122	120	124	123	103	81	66	63	
All-Sky Solar Radiation	RadAvg	1.86	2.90	3.81	4.75	5.49	6.11	6.61	5.44	4.27	2.81	1.86	1.40	
	RadStd	0.20	0.33	0.39	0.48	0.41	0.42	0.23	0.45	0.32	0.36	0.16	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-0.69	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page